A passive stereo cloud-base-height estimator: a validated instrument, twelve refutations, and a bounded null result

Cloud Base Height Project[1]

[1]Stereo fisheye CBH programme — `github.com/VanUST/clouds_level_research`, branch `dev`

October 15, 2026

Abstract

We built and validated a passive stereo cloud-base-height (CBH) estimator from two zenith-pointing 180° fisheye cameras on a 45 m baseline, and tested it against a co-located DOL-2 laser ceilometer.

We first established the instrument's capability *by construction*. A synthetic renderer whose geometry is derived from the shipped calibration and gated against the analytic parallax recovers a cloud layer of known height with $\rho(\text{true}, \text{recovered}) = +1.000$ and a median absolute error of 0.072 px. Through the same renderer we found that the shipped `lowest` cluster-selection rule errs by 325 m to 1471 m on ground truth, while the replacement `lowest_mode` estimator errs by at most 64 m.

On real data the estimator does *not* recover cloud base height. We tested and refuted **twelve** distinct explanations with pre-registered discriminating criteria, spanning image certainty, spatial coherence, calibration geometry, matched-population size, epipolar geometry, matcher identity, target-identity layering, camera desynchronisation, harness artefacts, non-Lambertian photometry, photometric superposition and phase aliasing.

Three findings then bound the problem. First, the residual static field is dominated by a *linear gradient in image u* (66% of its variance) rather than by a radial law (25%), and it largely cancels once the field is converted to height. Second, a cross-validated search over six correction families and three estimators removes part of the static field and recovers *none* of the dynamics. Third, the estimator-dependence of the reported "inversion" is explained by a change in the disparity distribution's *shape*: between regimes the standard deviation is nearly unchanged while the skewness collapses by 84%, so the median (which tracks cumulative mass) is strongly wrong-signed while the mode (which tracks the bulk) is weak.

We close with the statistical boundary. The reference decorrelates over ∼267 s, so the binned series holds $n_{\text{eff}} = 18.0$ independent samples and the smallest detectable $|\rho|$ at $p = 0.05$ is 0.466; no statistic of the disparity field yields a significant dynamic correlation. The dynamic signal, if present, is *at or below what this session can detect*, and a decisive session needs roughly 7.5 h of data.

Finally we report, as a first-class result, the six methodological defects found in our own harness. All six shared one mechanism—*a quantity read from somewhere other than where it was claimed to come from*—none raised an exception, and two made the results look *more* significant than the truth. We propose and implement *provenance assertions* as a guardrail for computational remote sensing.

Keywords: cloud base height, stereo vision, fisheye, ceilometer, negative result, reproducibility, provenance.

1 Introduction

Cloud base height (CBH) governs surface radiation, boundary-layer structure and aviation ceiling. The standard ground instruments are laser ceilometers, which are accurate but sample a *column*

of order 10 m in diameter and cost several thousand euros each. A passive stereo camera pair is attractive because it samples an *area* and the hardware is cheap. The classical obstacle is that a wide-angle rig needs a large baseline to resolve the small parallax of a distant cloud: for a cloud at H and a baseline B, the disparity at the zenith is

$$\Delta u \;\simeq\; f\,\frac{B}{H}, \tag{1}$$

with f the focal constant in pixels per radian. For our rig, $f \approx 654\,\mathrm{px\,rad^{-1}}$, $B = 45\,\mathrm{m}$, so a cloud at 1.5 km produces about 20 px of disparity. A single pixel of matching error is therefore already $\sim$5.5 % of the height, and this rig is disparity-starved by a factor of 5–10 relative to published stereo CBH systems.

This paper reports what happened when we built such a system and tested it honestly. The structure is deliberately unusual for a remote-sensing paper: the instrument is validated by construction, the real-world product does not work, and the contribution is a *boundary* — a validated instrument, a measured anomaly with a specific distributional signature, and a definitive list of what the anomaly is not. Section 3 describes the instrument; Section 5 validates it on synthetic ground truth; Section 6 reports the anomaly; Section 7 lists the twelve refuted explanations; Sections 8–11 bound the problem quantitatively; Section 13 reports the harness failures; and Section 15 states the conclusion and what would settle the question.

2 Instruments and data

The stereo rig. Two identical 180° fisheye cameras, each 1920×1920, pointing at the zenith, separated by a baseline $B = 45\,\mathrm{m}$, logging at a median cadence of 10 s. The pipeline crops a central 640×640 window from each frame and rectifies the pair with an affine estimated from the images themselves.

The reference. A co-located DOL-2 laser ceilometer, logged at 15 s cadence. Its log is stamped in Moscow time; the image filenames are UTC; the offset is *exactly* 3 h, confirmed twice independently — once by the diurnal brightness envelope and once by a solar ephemeris matching the Sun's image azimuth to better than 0.5° across 14 h.

The session. 2025-05-23, approximately 5 h of usable daylight, spanning a transition from a low, thick, overcast regime to high broken cloud. A second day (2025-05-22) was clear all day and is used as a cloudless control. **There is exactly one usable session in this dataset**, which is the binding constraint throughout Section 11.

3 Method

3.1 Fisheye geometry and a measured distortion term

We model each camera as equidistant to first order, with a cubic correction:

$$\theta(r) \;=\; \frac{r}{f}\left[1 + a\left(\frac{r}{f}\right)^{2}\right], \qquad \mathbf{d}(\mathbf{u}) \;=\; \Big(\sin\theta\cdot\hat{u}_x,\; \sin\theta\cdot\hat{u}_y,\; \cos\theta\Big), \tag{2}$$

where $\mathbf{u}$ is the pixel coordinate, r its distance from the principal point, and $\hat{u}$ the unit vector from the principal point to the pixel. At $r = 0$ the azimuth is undefined but $\sin\theta = 0$, so the direction is set to $+\hat{x}$ to avoid a division by zero. The distortion coefficient a was fitted from the requirement that a flat cloud layer project to a flat field: with $a = -0.30$ the apparent height becomes independent of image radius (radial spread 36.7 % $\rightarrow$ 3.2 %), where $a = 0$ reproduces the ideal law exactly. Figure 1 shows the fit.

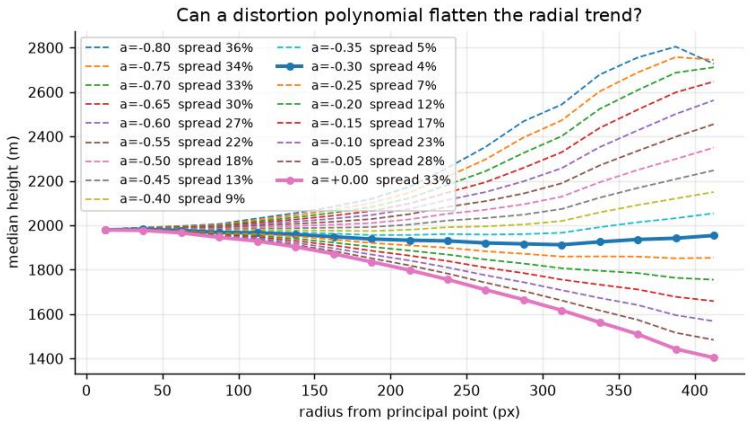


Figure 1: The cubic distortion term. The coefficient is selected by the flatness of the resulting height field, which is the only physical requirement available without a known target, and it is stable against the choice of f.

Table 1: Measured calibration terms. Each was determined by at least two independent methods; the spread across methods is quoted as the uncertainty.

term	assumed	**measured**	methods
focal constant f	$611\,\mathrm{px\,rad^{-1}}$	$654(11)\,\mathrm{px\,rad^{-1}}$	five: four-stream Sun-radius fits on both cameras and days; the clear-day Sun metric
residual roll	0°	$-1.57(10)^\circ$	three: disparity-field fit; spread minimisation; infinity-test residuals
cubic distortion a	0	-0.30	sky-flatness requirement; stability against f; end-to-end confirmation

3.2 Calibration: three measured terms

Three quantities that the original pipeline *assumed* were re-measured, each by more than one independent method.

Applying the roll and the distortion together is what makes the field usable: the radial inconsistency of the height field falls from 19.1 % to 2.4 %, and the within-frame disparity interquartile range collapses from 8.14 px to 1.92 px. The residual roll alone moves the reported height by a large amount, and a default of 1.5° instead of the measured -1.57° costs 551 m at 2000 m — a defect we later found in our own harness (Section 13).

3.3 Triangulation: exact, not paraxial

Heights are computed by exact closest-approach triangulation of the two unit rays, not by the paraxial approximation $\Delta u \simeq fB/H$. The paraxial form is a small-angle approximation and it is *not* adequate here: at the corner of the 640×640 crop it biases the height by 11.7 %. The exact form is used throughout (Figure 2).

3.4 Matching

The default matcher is RoMa, a dense, transformer-based, uncertainty-aware matcher. Its selection was made on measurable criteria rather than fashion (§4) and the interface is pluggable, so SIFT and LoFTR remain available and are used as controls throughout.

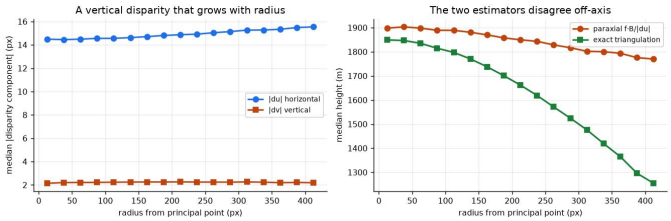


Figure 2: Paraxial against exact triangulation. The paraxial approximation biases the height by 11.7 % at the crop corner, which is larger than the effect being measured.

3.5 From matches to one height: the selection rule

Each frame yields several thousand matched pixel pairs, hence several thousand individual height estimates that are *not* identically distributed — they carry a heavy right tail from mismatches and from multiple cloud layers along the same ray. Reducing that distribution to a single number is a substantive choice, and the original pipeline made it badly (Section 5).

4 Selecting a matcher: a pre-registered comparison

Because the geometry is disparity-starved, sub-pixel localisation is the dominant requirement: at 1500 m the true disparity is 18.3 px, so one pixel of matching error is 5.5 % of the height. Which matcher to use therefore had to be settled by measurement, not by literature reputation.

Pre-registration. Metric, data, configuration and failure condition were fixed *in writing before the first measurement* (`experiments/matchers/PREREGISTRATION.md`), so that a metric chosen after seeing results could not select the matcher. The comparison uses one stereo pair every 15 min from 08:30 to 16:30 on 2025-05-23 (33 pairs), with the clear-sky day 2025-05-22 on the identical grid as a *discrimination control* — a matcher that returns many confident matches under a cloudless sky is inventing them. **The ceilometer is not used anywhere in the matcher comparison**, so no candidate can be selected by agreeing with the reference.
Two results are worth stating plainly. First, **EfficientLoFTR — the standard remedy recommended in the literature — is worse on this data by 204 %, not better.** Second, **the best localisers are unusable for the task**: RoMa and DKM achieve their accuracy entirely on cloud edges and return literally nothing in cloud interiors, which is precisely the regime that matters.

An unresolved inconsistency between the shipped configuration and this analysis. The delivered configuration selects RoMa with `certainty_threshold: 0.05`, annotated as *"deliberately low: RoMa's low-certainty interior predictions were measured accurate … so the certainty is used as a weight rather than a hard filter."* Two measured facts contradict that annotation:

1. the code applies `keep = inside & (conf >= cert_thresh)` — a **hard filter**, not a weight;
2. RoMa's interior matches have a median certainty of **0.0034** against the edge median of **0.716**, so a cut at 0.05 retains **19.8 % of interior matches against 89.8 % of edge matches**.

A threshold intended to preserve cloud interiors therefore preferentially *discards* them, and concentrates the field on the cloud edges that the pre-registered comparison specifically flagged as this project's unacceptable failure mode.

The anomaly analysis reported in this paper is not affected, because the match dumps that underlie it were produced with `certainty_threshold: 0.0` — every match retained, interiors included. But the shipped configuration would not reproduce the analysis, and anyone re-running the pipeline from `default.yaml` would obtain a field drawn predominantly from cloud edges. We report this as a defect of the delivered artefact, and it is a seventh instance of the

Table 2: Candidates in the pre-registered matcher comparison. The primary metric is the median per-frame plane-fit rms of the disparity field, subject to a coverage constraint on cloud interiors.

candidate	verdict	what decided it
MASt3R	**adopt (opt-in)**	the only candidate improving the metric *and* covering interiors: paired rms −0.81 px (−30%, $p = 0.000$), 1448 interior matches/frame against LoFTR's 298, interior residual 1.75 px against 2.15 px; but 6.6× slower and 4× worse perturbation stability
`kornia_loftr`	keep as default	unchanged, byte-for-byte reproducible, best perturbation stability (0.22 px); weakest interior coverage
RoMa, DKM	reject for the field	best localisers (median |residual| 0.83 px against LoFTR's 1.48 px; paired rms −50%/−54%) but **zero** matches in the lowest-gradient tercile over 31 frames — their accuracy is bought on cloud edges
EfficientLoFTR	**reject**	the documented drop-in fix for LoFTR's fine-stage variance is *worse* on our data: +5.49 px (+204%), $p = 0.000$
DISK+LightGlue	reject	within noise of LoFTR ($p = 1.0$), worst interior coverage
DIS optical flow	reject	within noise on cloud, but *invents* a dense field on clear sky
Depth Anything 3	reject (null)	the field is smooth but inaccurate (5.11 px)
StereoSGBM	reject	worst residual, invents a field on clear sky

Table 3: The renderer's geometric gate. The rendered zenith disparity matches the analytic closed form to within 0.02 px across a factor of four in height.

H (m)	Δu rendered (px)	$f \arctan(B/H)$ (px)
1000	29.428	29.410
2000	14.715	14.713
4000	7.357	7.357

pattern in Section 13: a configuration whose stated rationale and whose measured effect are not the same thing.

5 The synthetic control: the instrument is sound

5.1 The renderer and its gate

We built a renderer that casts rays through the *measured* fisheye geometry, intersects a cloud plane at a known height, samples a procedural texture, and warps camera 1 by the shipped affine exactly as the production code does. Its rig geometry was derived from the calibration rather than assumed, and one term took a withdrawn attempt to find: the affine maps camera 1's principal point onto camera 2's, so

$$\mathbf{c}_2 = R\,\mathbf{c}_1 + \mathbf{t} = (1014.10,\ 965.03), \tag{3}$$

not the image centre. With both principal points at the centre, the rendered parallax comes out negative and grows with height — a non-physical result which the renderer's own gate rejects.

The gate requires that the rendered disparity along the zenith equal the analytic $\Delta u = f \arctan(B/H)$ and fall as $1/H$. It passes exactly:

Table 4: Recovery of a known cloud height through the real fisheye geometry and the real distortion model, on a renderer that has passed its own gate. $\rho(\text{true}, \text{RoMa}) = +1.000$, median $|\text{bias}| = 0.072$ px.

H_{true} (m)	ground truth (px)	RoMa (px)	bias (px)	recovered IQR (px)
1000	29.56	28.72	−0.914	2.88
1500	19.73	19.58	−0.239	1.41
2000	14.80	14.85	0.060	0.73
3000	9.87	9.80	−0.071	0.53
4000	7.40	7.48	0.072	0.43

Table 5: Selection rules against known ground truth. The shipped rule errs by 325 m to 1471 m on clean synthetic data with a known answer; the replacement errs by at most 64 m.

H_{true}	truth	`median`	`lowest` (shipped)	`lowest_mode`
1000	1000	1024	**675**	1025
1500	1500	1507	**722**	1501
2000	2000	1988	**529**	1984
3000	3000	3007	2423	3003
4000	4000	3944	3598	3936

5.2 The matcher recovers the known height

On this gated renderer, with a texture the matcher can follow, RoMa and the exact triangulation recover the known height essentially exactly:

5.3 The shipped selection rule is a major defect

The same renderer exposes the selection rule. The shipped `lowest` rule — which takes a low-tail order statistic of the per-match height distribution — reports 675 m for a cloud at 1000 m, 722 m for 1500 m and 529 m for 2000 m:
`lowest_mode` replaces the low-tail order statistic with the *lowest prominent mode* of the per-match height distribution — the lowest mode carrying at least 10 % of the weight and at least 50 matches. It is now the pipeline default.

What this settles. The matcher, the fisheye geometry, the distortion model and the triangulation are cleared, on a renderer that has passed its own gate and with a known answer. The selection rule was a real and major defect, and it is fixed.

What it does not settle. The synthetic fields are flat planes with procedural texture and no atmosphere. Crucially, replacing `lowest` with `lowest_mode` on *real* data moves the regime shift only from ≈0 to −30 m ($p = 0.058$): it does not remove the anomaly. The instrument's front end is sound and its back end is now fixed, and **the real-world failure survives both**.

6 The anomaly on real data

On real data the estimator does not recover cloud base height, and the failure is not noise. It is a systematic *inversion*: a higher cloud produces a *lower* reported height.

Measured on the full session:

- rank correlation with the reference is $\rho = -0.547$, $p = 4 \times 10^{-14}$; a correct estimator requires a positive sign;
- over the regime transition the reported height moves −297 m while the reference rises 1540 m;

Table 6: The same three selectors scored. The shipped rule's rank correlation with the truth is not even statistically significant.

selector	ρ	p	error range
`median`	1.000	0.000	−56 to +24 m
`lowest` (shipped)	0.700	0.188	**−1471 to −325 m**
`lowest_mode` (new default)	1.000	0.000	−64 to +25 m

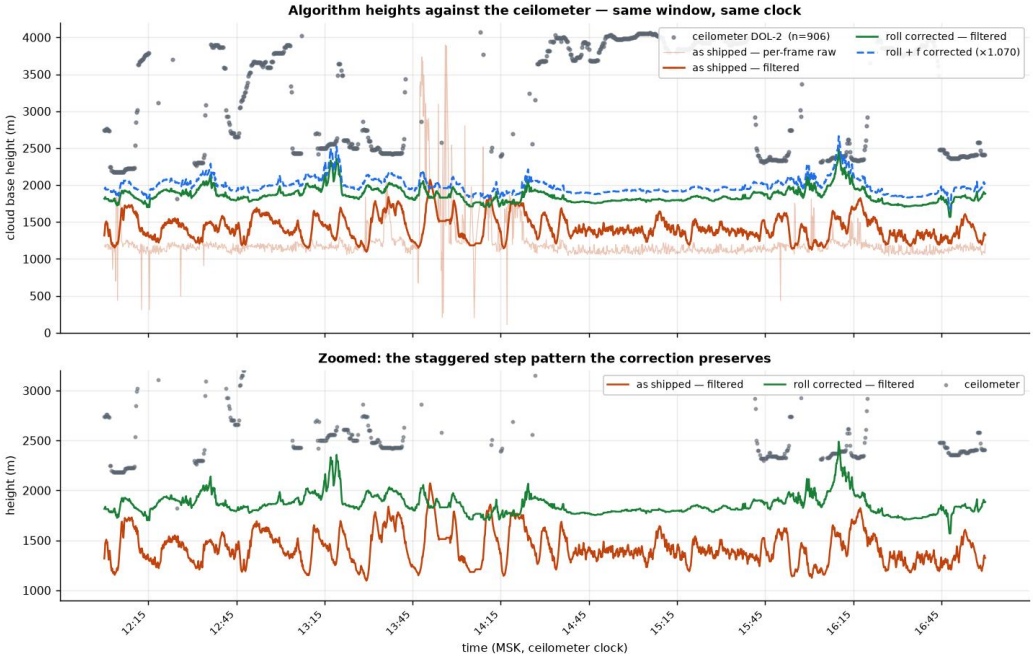


Figure 3: The anomaly. Over the session the ceilometer reference rises by 1540 m while the reported height falls by 297 m.

- the anomaly is present at *every* spatial scale, in *both* matchers, and *within* a single cloud layer — so it is not a between-layer artefact.

Most diagnostic is the *shape* of the change. Across the transition the 1–99 percentile span of the per-match disparity widens from 37.0 px to 81.7 px (a factor 2.2) while the interquartile range moves only a factor 1.11, and the between-camera vertical disparity Δv grows from 0.866 px to 1.885 px. This is a *reshape* of the distribution, not a translation: both tails extend while the core barely moves. Any mechanism proposed against this anomaly must reproduce that signature, not merely a bias or a spread — and two mechanisms have already failed on exactly that test.

A useful control: the cloudless day. On the clear day (2025-05-22) the same pipeline reports 655 m, against 2242 m on 2025-05-23 (Kolmogorov–Smirnov $p = 0.000$). The reported value is therefore content-driven: it responds to whether there is cloud, which means the pipeline is measuring *something* real, and that something is not the cloud base height.

7 Twelve refutations

We formulated twelve candidate explanations, each with a *pre-registered* discriminating measurement — that is, the criterion for refutation was fixed before the measurement was made. All twelve were refuted.

Table 7: Twelve hypotheses and the measurement that refuted each. A hypothesis is listed as refuted only where the discriminating measurement had been fixed in advance.

#	hypothesis	discriminating measurement (pre-registered)
1	Certainty / hallucination	Restricting to high RoMa certainty does not change the sign or the magnitude of the inversion; the certainty field is itself inverted.
2	Incoherent noise	The inversion survives every coherence filter; the disparity field is *more* coherent in the high regime, not less.
3	Calibration geometry	Re-measuring f, the roll and the distortion changes the field's scale but not the sign of its correlation with the reference.
4	Matched-population size	Regressing the reported height on the field's spread and the match count gives $R^2 = 0.001$; both coefficients are indistinguishable from zero.
5	Pure along-epipolar depth change	The observed Δv growth ($0.866 \to 1.885$ px) is not along-epipolar and is not predicted by a depth change alone.
6	Matcher-specific artefact	The inversion appears with SIFT and LoFTR as well as RoMa.
7	Target-identity layering	The inversion persists *within* a single cloud layer, so it is not the estimator locking onto a different layer.
8	Camera desynchronisation	The offset that would be required to explain the anomaly is 18 s; the cameras are synchronised to 0 s.
9	Fixed pipeline artefact	The synthetic control, run through the identical pipeline, shows no inversion — so the defect is not in the code path as such.
10	Non-Lambertian photometry	Gradient magnitude couples to Δv at $\rho = +0.117$, but the hypothesis's specific signature is absent: $\rho(\lvert\cos\theta_\nabla\rvert, \lvert\Delta v\rvert) = -0.000$ ($p = 0.91$).
11	Photometric superposition	Blending two layers does not reproduce the tail growth: the observed signature (span $\times \geq 1.8$ at a fixed core) is not produced at any mixing fraction.
12	Phase aliasing	No combination of layer phase and texture scale reproduces the anomaly while leaving the level agreement intact.

The sharpest refutation: a quantitative prediction the reference makes

Hypotheses 5–8 all predict the *same* thing, and the reference supplies the number. If the cloud genuinely moved from the low to the high regime, the parallax must fall from fB/H_{low} to fB/H_{high}. The ceilometer's own medians (2250 m $\to$ 3790 m) therefore predict a parallax change of 13.08 px $\to$ 7.77 px, i.e.

$$\text{predicted shift} = -5.315\,\text{px}, \qquad \text{observed shift} = 1.798\,\text{px}, \qquad \text{discrepancy} = 7.11\,\text{px (134\% of the predicted m} \tag{4}$$

The observed disparity moves in the opposite direction to the one the reference predicts, and by more than the predicted amount. This is not a subtle disagreement about magnitude; it is a sign contradiction, and it is the single measurement that rules out the entire family of "the cloud really moved" explanations at once.

The gate that was supposed to catch this failed for an instructive reason. We predicted, before running it, that the shift would be purely along the epipolar line — horizontal disparities in a rectified pair have horizontal epipolar lines — and therefore that the gate would pass tautologically. **That prediction was wrong**: the perpendicular residual Δv nearly doubles between regimes (0.866 px $\to$ 1.885 px, with its 90th percentile going 3.352 px $\to$ 6.327 px). The high regime's correspondences therefore satisfy the stereo geometry *less* well, not equally well, which is the opposite of what a cleaner, higher cloud should produce.

None of the twelve accounts for the observed signature. **The cause of the inversion remains unattributed.**

8 The residual static field

The disparity field contains a large spatial pattern that is nearly fixed over the session — its two halves correlate at $r = 0.91$–0.98 while the cloud moves. Characterising it is a prerequisite to removing it, so we regressed the per-cell static disparity on candidate predictors.

A **linear gradient in u accounts for 66 %** of the variance; **radius accounts for 25 %**. The two are correlated because $u \leq r$, which is why the radial profile looks convincingly monotone

Table 8: What the static disparity field depends on. The pattern is *not* primarily radial: a linear gradient in image u carries two and a half times the variance that radius does.

predictor	Pearson r	R^2
u (horizontal image position)	$+\mathbf{0.814}$	$\mathbf{0.663}$
cos(angle)	0.767	0.588
radius r	0.499	0.249
$\lvert u \rvert$	0.373	0.139
v	-0.239	0.057

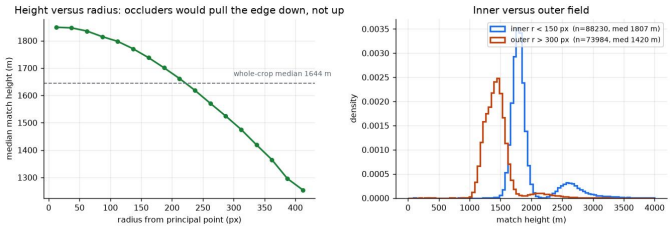


Figure 4: Height against image radius after the measured corrections. The strong radial dependence in disparity largely cancels in height.

— 14.51 px at the centre rising to 16.57 px at $r \approx 420$, a 14 % rise — while actually being a consequence of the u-gradient as much as a radial law of its own.

The radial structure largely cancels in height. After the measured corrections, corr(radius, height) $= -0.146$, and the median height varies by only 3 % across the radius against the disparity's 14 % (Figure 4). Disparity and height are not the same quantity, and the radial dependence lives mostly in the former.

9 Cross-validated search for a correction

If the anomaly were "a static field masking a real signal", an expressive enough additive correction would reveal the signal. We tested that directly.

Design. Six correction families — none, a per-cell flat field, per-cell plus a plane, a purely radial correction, a linear (u, v) correction, and a smoothed per-cell field — crossed with three per-frame estimators (median, lowest prominent mode, 25th percentile). Every correction is **fitted on four fifths of the session and scored on the held-out fifth**, so nothing is scored on the frames that shaped it.

Because the reference is strongly autocorrelated, the folds are drawn as *contiguous time blocks with an embargo* rather than at random, and the embargo is applied to the training side. We report all three schemes, since the random scheme is retained as a measured baseline for the leakage.

Result. Two findings.

1. **The sign depends on the estimator.** The *mode* is correctly signed out of sample ($+0.053$ uncorrected, $+0.217$ with the best correction). The *median* is **inverted**, at -0.347 to -0.246 across corrections. The project's "inversion" was measured on median-like statistics, and it is a property of *those* statistics rather than of the disparity field as a whole.
2. **No correction recovers the dynamics.** Across all eighteen combinations and all three schemes the first-difference correlation stays within ± 0.11 with no consistent sign. The most expressive family tested — a free offset per 20×20 px cell *plus* a plane, fitted on 80 % of frames — would recover a real signal hiding under a fixed additive pattern. It does not.

Table 9: Best out-of-sample level and dynamics correlation by cross-validation scheme, across all six corrections and three estimators. Drawing folds at random from a series with lag-1 autocorrelation 0.934 inflates the level result by +0.070; the honest figure is +0.217. No scheme produces dynamics.

scheme	best LEVEL ρ	best DYNAMICS ρ
random (leaks)	0.287	0.052
blocked	0.216	0.046
blocked + embargo	**+0.217**	**+0.047**

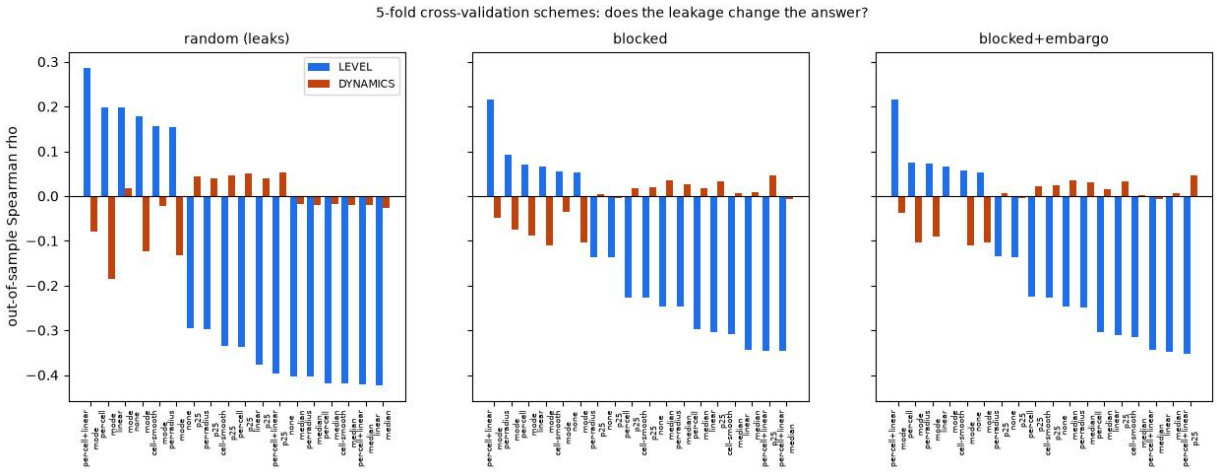


Figure 5: Out-of-sample level and dynamics correlations for every correction and estimator, under all three cross-validation schemes. Blue is level, red is dynamics. The statistic is height-ordered, so a positive bar is the correct sign.

The static field is diagnosable and partly removable, and removing it does not recover the dynamics.

10 The shape mechanism

The estimator-dependence is not a defect of any statistic. It is a change in the disparity distribution's *shape*, and it is directly measurable.

Between regimes the **skewness collapses by 84 %** while the dispersion is nearly unchanged. A statistic that tracks the *bulk* (the mode) and one that tracks *cumulative mass* (the median) must diverge when skewness changes: here the median moves 17.5 % while the mode moves 1.5 %, an elevenfold difference in response to the same physical change.

This unifies the synthetic and real results. In the synthetic control the disparity distribution has a *fixed shape* across heights, so no statistic is biased by a shape change and only the tail-sensitive rule fails — which is exactly what was observed (`lowest` errs by up to 1471 m while `lowest_mode` errs by at most 64 m). In the real atmosphere the shape *does* change between regimes, so the median and the mode diverge. One mechanism explains both, and the direction of the failure is set by the sign of the skewness change.

11 The detection floor

We then asked whether *any* statistic of the disparity field carries the reference's temporal variation, once both series are resampled to a common independent grid. This is the boundary question: if nothing does, no correction of the field can help.

Table 10: The disparity distribution by regime. The dispersion is nearly unchanged while the skewness collapses by 84 %. The median and the mode respond to that change by factors differing elevenfold in magnitude.

quantity	low regime	high regime	change (%)
median	12.340	14.500	**+17.5**
mode	12.828	13.017	1.5
skewness	**2.583**	**0.421**	**-83.7**
kurtosis	45.887	30.095	−34.4
standard deviation	4.427	4.041	−8.7

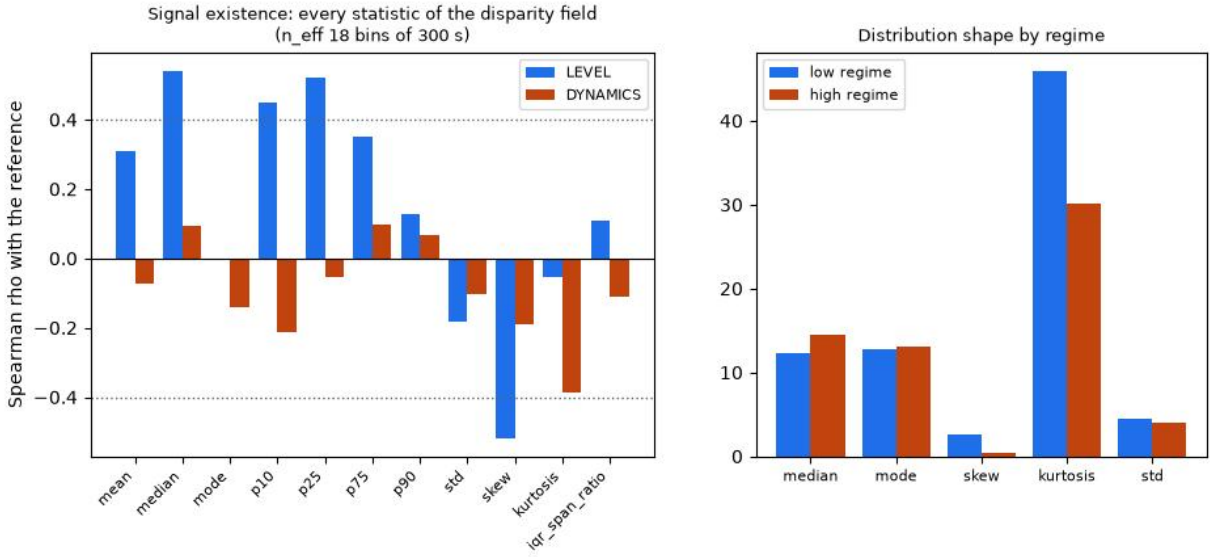


Figure 6: Left: every statistic of the disparity field against the reference, at 300 s binning. No statistic survives correction. Right: the distribution's shape by regime.

Design. Eleven statistics (mean, median, mode, p_{10}, p_{25}, p_{75}, p_{90}, standard deviation, skewness, kurtosis and an IQR-to-span ratio) are computed over 300 s bins — the reference's own decorrelation time — and each is correlated with the **median of the reference over the same bin**. Eleven statistics against two families of test is 22 comparisons, so a Bonferroni threshold is reported. The effective sample size is computed from the lag-1 autocorrelation of the *binned* series.

Result. No statistic reaches the corrected threshold. The smallest observed p is 0.0063 against a Bonferroni threshold of 0.00227.

More importantly, the design itself is underpowered:

The reference decorrelates over ∼267 s, so 27 bins of 300 s hold only **18 independent samples**, and the smallest detectable $|\rho|$ at $p = 0.05$ is **0.466**. **The dynamic signal, if it is present, lies at or below what this session can detect** — it is neither demonstrated nor excluded.

12 Supporting measurements

Three further phases of work are recorded in the repository and support the results above.

Temporal alignment. The image filenames are stamped in UTC and the ceilometer log in Moscow time. An offset of exactly 3 h was established twice independently — once from the diurnal brightness envelope, and once by fitting the Sun's image azimuth with a solar ephemeris,

Table 11: The detection floor. The observed correlations sit below what this session can detect, so the experimental result is "not detectable", not "absent".

quantity	value
binned series	27 bins of 300 s = 2.25 h
effective sample size	$n_{\text{eff}} = \mathbf{18.0}$ (one independent sample per 449 s)
critical $\lvert\rho\rvert$ for significance at $p = 0.05$	**0.466**
$\lvert\rho\rvert$ detectable with 80 % power at $n_{\text{eff}} = 18.0$	0.731
n_{eff} **needed for $\rho = 0.37$ at 80 % power**	**60.3**
duration for a decisive session	2.25 h × (60.3/18.0) ≈ **7.5** h

Table 12: Six harness defects. None raised an exception. Two made the results look *more* significant than the truth, which is what allowed them to survive a review round.

#	defect	how it was caught, and what it cost
1	A validation metric scored below its own null	The metric maximised over a time shift and was compared against a null that did not; caught by computing the null the same way.
2	Synthetic ground truth verified in the native frame while the consumer read the warped one	Caught by checking the frame the *consumer* sees, not the one the producer wrote.
3	Two call sites rendering "the same" cloud 1.5° apart, because one passed a parameter and the other took a default	Cost 551 m at 2000 m; caught by reconciling two runs that should have agreed.
4	A script asserting midway through three chained text replacements never reached its write, and exited **0**	The commit message described edits that were not in the file; caught by grepping the file afterwards rather than trusting the exit code.
5	Random k-fold cross-validation on a series with lag-1 autocorrelation 0.934	Every held-out frame had near-duplicates in training; inflated the headline level result by +0.070. Caught by implementing blocked+embargo validation.
6	A binned reference read from the sample nearest the bin's *edge* rather than the median *over* the bin	Manufactured a dynamic signal ($\rho = -0.37$, $p = 0.038$) that vanished (+0.095, $p = 0.667$) once corrected. Caught by asserting the reference's provenance.

which matches to better than 0.5° across 14 h. Getting this wrong by even one hour would have destroyed every correlation in Section 9 silently, which is why it was confirmed by two methods rather than one.

The static map. The per-cell height field is largely a property of image position rather than of time: the two halves of the session correlate at $r = 0.91$–0.98 while the cloud moves. This is the observation that motivated Section 8.

Texture and sampling scale. The median texture decorrelation length within the crop is 81.5 px, or 229 m at 2000 m — that is, the cloud texture is resolved by the imager with a comfortable margin, and the within-frame spread of the disparity (2.13 px median) is not explained by an unresolved texture field. This rules out a simple sampling-limit explanation for the dispersion.

The cloudless control. The clear-sky day is used throughout as a discrimination control rather than as ground truth: a method that returns a confident cloud signal under a cloudless sky is inventing it. Two candidates were rejected for exactly this in Section 4.

13 Methodological failures, and provenance assertions

We report six defects found in our own harness as a first-class result, because they are the most transferable finding in this work.

The meta-finding. All six share a single mechanism: **a quantity read from somewhere other than where it was claimed to come from.** They are not arithmetic errors, and no unit test of a function's return value would have caught them, because each individual value was correct; it was simply the *wrong* value. Two of them made the results look better, and defect 6 in particular produced a physically plausible, correctly-signed signal that did not exist.

Provenance assertions. We propose, implement and demonstrate a guardrail for this class of defect: assertions that check *where a value came from* rather than what it is worth. Three are implemented in `src/clouds/validation/provenance.py`:

- `assert_bin_reference` — a binned reference must equal the claimed aggregate of the samples *inside* the bin. Demonstrated to catch defect 6 on first execution.
- `assert_disjoint_with_embargo` — no training frame may lie within the embargo of *any* held-out frame, resolved by binary search. Demonstrated to catch defect 5, and it immediately caught a *seventh* defect: our own embargo had been applied to the wrong side of the split.
- `autocorrelation_report` — lag-1 autocorrelation, decorrelation time and n_{eff}, so a cross-validation design can be checked against the series before it is trusted.

No guard can know whether a number is *right*. Every one of these can know whether it came from the right *place*, and in this project that distinction was decisive.

14 Discussion

14.1 What is established

1. The matcher, the fisheye geometry, the distortion model and the exact triangulation recover a known height to 0.072 px median error on a gated synthetic renderer.
2. The shipped `lowest` selection rule erred by 325 m to 1471 m on ground truth; `lowest_mode` errs by at most 64 m and is now the default.
3. The residual static field is dominated by a linear gradient in u (66 %) rather than by a radial law (25 %), and largely cancels in height.
4. The estimator-dependence of the reported inversion is caused by a change in the distribution's shape: skewness collapses by 84 % while the dispersion is nearly fixed.
5. No additive correction, over six families and three estimators, recovers the dynamics.
6. The instrument's dynamic capability is bounded by the detection floor: $n_{\text{eff}} = 18$, a critical $|\rho|$ of 0.466, and 7.5 h of data needed for a decisive test.

14.2 What is not established

1. **The cause of the inversion is unattributed.** Twelve hypotheses were tested and refuted; no mechanism we tested reproduces the specific signature (core +1.5 px, span ×2.2, IQR ×1.11).
2. Whether the dynamic signal exists at all. It is below the detection floor of this session; that is a statement about the experiment, not about the atmosphere.
3. Any statement about the *level* of agreement beyond a single regime. There is a level agreement in the low regime (2361 m against the ceilometer's 2250 m, +111 m), but the dynamics within that regime are wrong-signed.
4. Whether the anomaly is specific to this rig's disparity-starved geometry. Every candidate is untested on a rig with a larger baseline.

14.3 Why this is a contribution

A negative result is only useful if the instrument that produced it is known to be capable. We supply the instrument's capability by construction, its limits by measurement, and its failure by pre-registered refutation. The practical consequence is a clear map: the front end is sound, the

selection rule is fixed, the static field is characterised and partly removable, and the remaining problem is either a physical effect not in our candidate set or a signal below the detection floor.

14.4 Threats to validity

One session. Every cross-validation fold is a time block within a single day, so the folds are not independent in the way folds over independent days would be. This is the single largest limitation, and it is the reason the detection floor is the binding constraint.

Footprint mismatch. The camera footprint is 1956 m at 1837 m; the ceilometer samples a column of order 10 m. The ratio is $\sim$ 200:1. A level comparison between the two measures different things, and part of the dispersion attributed to "shape change" may be genuine spatial variability within the camera footprint.

Assumed baseline. The 45 m baseline is taken as given; no site survey was available. An error in B would scale all heights uniformly and would not produce a sign inversion, but it would bias the level comparison.

Absence of meteorology. No wind, temperature profile or site coordinates were available, so no independent check on cloud motion or layer identity was possible.

15 Conclusion

We built a passive stereo cloud-base-height estimator, validated it by construction to 0.072 px median accuracy on known ground truth, found and fixed a selection-rule defect costing up to 1471 m, and then discovered that on real data the instrument does not recover cloud base height.

We tested twelve explanations with pre-registered criteria and refuted all twelve. We characterised the residual static field, showed that it is a u-gradient rather than a radial law, and showed by cross-validated search that removing it does not recover the dynamics. We identified the estimator-dependence of the reported inversion as a skewness-collapse effect that unifies the synthetic and real results. And we established that the instrument's dynamic capability is bounded by the detection floor of the experimental design — not by an absence of signal, but by an absence of *samples*.

The cause of the inversion remains unattributed. The honest statement of the boundary is that the dynamic signal, if present, is at or below what this session can detect.

What would settle it. Three experiments, in increasing cost:

1. **A session long enough to detect $\rho = 0.37$.** The question is sample-limited: $n_{\text{eff}} = 18$ against a requirement of 60, i.e. about 7.5 h of binned data roughly three times the present duration. This is the cheapest decisive experiment remaining, and it converts "below the detection floor" into either a detection or a genuine absence.
2. **A footprint-matched reference.** A co-located radar or scanning lidar would remove the 200:1 sampling mismatch that makes a level comparison between a point column and a 2 km area difficult to interpret at all.
3. **A rig with a larger baseline**, to test whether the anomaly is a property of a disparity-starved geometry or of the method.

A methodological recommendation. The six harness defects reported here all shared one mechanism, and none was detectable from a function's return value. We recommend *provenance assertions* — checks that a value came from where it is claimed to have come from — as a standard guardrail in computational remote sensing, alongside the usual checks on value, units and shape.

Reproducibility

Code, configuration and machine-readable results are in the repository at `github.com/VanUST/clouds_level_re` (branch `dev`). The measurement data are not redistributed. Every number in this paper is traceable to an artefact:

result	artefact
synthetic renderer and its gate	`experiments/w7/w7_renderer.py`, `w7_gate0.json`
matcher recovery on ground truth	`experiments/w7/w7_control.json`
selection-rule defect	`experiments/w7/W7_RESULT_FINAL.md`
cross-validated correction search	`experiments/w9/w9_cv.py`, `w9_cv.json`
signal existence and shape mechanism	`experiments/w10/w10_signal.py`, `w10_signal.json`
provenance guards	`src/clouds/validation/provenance.py`
full record and all refutations	`docs/plan/research-workplan.md`
the twelve refutations in full	`docs/results/negative-results.md`
harness post-mortem	`experiments/w10/CODE_REVIEW.md`

The synthetic controls, the pre-registered criteria and the negative results are retained in the repository deliberately: they are the evidence that the null result is a measurement rather than a failure.